

Accounting and Budgetary Systems for Engineers - INFO7225

MGEN

Chapter 4. Job Costing System

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Chapter 9

Accounting Information Systems and Business Processes: Part II

Accounting Subsystem.

Because the cost of goods sold is likely to be the largest expense on a manufacturing firm's income statement, a critical part of the production process is an AIS's cost accounting subsystem. The cost accounting subsystem provides important control information (e.g., variance reports reflecting differences between actual and standard production costs) and varies with the size of the company and the types of product produced. As you might guess, a bakery producing baked goods would collect very different data in its AIS compared to an automobile manufacturer. Cost accounting subsystems for manufacturing organizations are commonly job costing, process costing, or activity-based costing systems.

A **job costing information system** keeps track of the specific costs for raw materials, labor, and overhead associated with each product or group of products, called a “job.” This type of costing system is most appropriate for manufacturers of large-scale or custom products, such as home builders or book publishers. Manufacturers of homogeneous products (such as soft drinks or toothbrushes) that are produced on a regular and continuous basis use a **process costing information system**. In this system, it is not feasible or practical to keep track of costs for each item or group of items produced. Instead, process costing systems use averages to calculate the costs associated with goods in process and finished goods produced.

Activity-based costing systems help managers describe processes, identify cost drivers of each process, and then determine the unit costs of products created in each process. By studying their business processes, managers are in a better position to recognize opportunities to improve those processes. Thus, activity-based costing gives managers a better understanding of their processes, an improved ability to allocate indirect costs to those processes, and a better understanding of the true cost of each product. The systems flowchart in **Figure 9-7** shows a



Job Order Costing

LEARNING OBJECTIVES

After completing this segment, you should be able to

- Determining the costs of an individual job using job order costing;
- Determine and dispose of under-applied or over-applied overhead;
- Prepare journal entries for a job order cost system



Job Order Costing: Determining Primary Costs

(Ending)

	Raw Materials Inventory: Vinyl	
Beginning inventory	1,000	
4/2/2017 (Purchases)	10,000	
Balance	11,000	

(Ending)

	Raw Materials Inventory: Black Ink	
Beginning inventory	300	
4/2/2017 (Purchases)	500	
Balance	800	

(Ending)

	Raw Materials Inventory: Red Ink	
Beginning inventory	300	
Balance	300	

(Ending)

	Raw Materials Inventory: Gold Ink	
Beginning inventory	300	
Balance	300	

	Raw Materials Inventory: Grommets	
Beginning inventory	120	
Balance	120	

	Raw Materials Inventory: Framing Wood	
Beginning inventory	60	
Balance	60	

Job Order Costing: Product Costs Traced to Each “Job”

MATERIALS REQUISITION FORM

Dinosaur Vinyl, Inc.

Material Requisition No.: 3392

Job No.: MAC001

Date of Request: ~~4/5/2017~~

Date Needed: ~~4/5/2017~~

Description	Quantity	Unit Cost	Total Cost
Raw materials inventory: Vinyl	1	\$300	\$300
Raw materials inventory: Black ink	2	50	100
Raw materials inventory: Red ink	1	60	60
Raw materials inventory: Gold ink	1	60	60
Raw materials inventory: Grommets	12	10	120
Raw materials inventory: Framing wood	40	1.50	60
			\$700

Requested by: John Ming

Date: ~~4/5/17~~

Authorized by: Jula Clark

Date: ~~4/5/17~~

Direct Materials Needed for Job MAC001

Item	Units	Cost per Unit	Item Cost	Total Cost
Production Department (4/2/2017)				
Vinyl	1	\$ 300	\$300	
Black ink	2	50	100	
Red ink	1	60	60	
Gold ink	1	60	60	\$520
Finishing Department (4/14/2017)				
Grommets	12	\$ 10	\$120	
Framing wood	40	1.50	60	180
Total Direct Materials				\$700

	Raw Materials Inventory: Vinyl	
Beginning inventory	1,000	
4/2/2017	10,000	
4/2/2017		300
Balance	10,700	

	Raw Materials Inventory: Grommets	
Beginning inventory	300	
4/14/2017		120
Balance	180	

	Raw Materials Inventory: Black Ink	
Beginning inventory	300	
4/2/2017	500	
4/2/2017		100
Balance	700	

	Raw Materials Inventory: Framing Wood	
Beginning inventory	300	
4/14/2017		60
Balance	240	

	Raw Materials Inventory: Red Ink	
Beginning inventory	300	
4/2/2017		60
Balance	240	

	Raw Materials Inventory: Gold Ink	
Beginning inventory	300	
4/2/2017		60
Balance	240	

**Work-in-process
Inventory**

Direct Labor for Job MAC001

Item	Hours	Rate per Hour	Item Cost	Total Cost
Production Department (4/5/2017)				
Material Handler	1	\$15	\$15	
Print Technician	1	15	15	\$30
Finishing Department (4/15/2017)				
Production Assistant	1	\$18	\$18	
Production Assistant	1	18	18	<u>36</u>
Total Direct Labor				\$66

Costs Accounted for in the Work in Process Inventory

Work-in-process Inventory

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Job Order Costing Record Keeping: Job Cost Sheet

JOB COST SHEET
Dinosaur Vinyl, Inc.

Job No.: ~~5416~~ MAC001

Customer No.: 2501723

Customer: Macs & Cheese

Date Started: ~~Dec. 22, 2018~~

Units Ordered: 1

Date Completed: ~~Feb. 22, 2019~~

Direct Material	Units	Price	Amount
Vinyl	1	\$300	\$ 300
Black printing ink	2	50	100
Red printing ink	1	60	60
Gold printing ink	1	60	60
Grommets	12	10	120
Framing wood	40	1.5	60
Total Direct Materials			\$ 700

Direct Labor	Hours	Wage Rate	Amount
Material Handler	1	\$ 15	\$ 15
Print Technician	1	15	15
Production Assistants	2	18	36
Total Direct Labor			\$ 66

Manufacturing Overhead	Base Units	Rate	Amount
Direct Labor Cost			
Total Manufacturing Overhead			
Total Job Cost			
Total Revenue			

**Job Order
Costing Record
Keeping:
Individual “Job
Cost Sheets”**



Job Order Costing: Determining Manufacturing Overhead Cost (Estimated)

Computing a Predetermined Overhead Rate

Estimate MOH costs:

... uses the expenses from the prior two years to estimate the **overhead costs** for the upcoming year to be **\$250,000**:

Manufacturing overhead

Annual Estimate

Indirect labor	\$ 5,000
Indirect materials	20,000
Utilities	75,000
Depreciation	90,000
Insurance	35,000
Interest expense	25,000
	<u>\$250,000</u>

Find a cost driver:

... also used its payroll records to estimate that it will spend **\$100,000** on **direct labor**.

Using the predetermined overhead rate calculation, the overhead rate is:

Estimated (budgeted) Overhead Cost (\$250,000)

Expected (budgeted) Level of Activity (\$100,000)

Journal Entry to Record Overhead Costs for Job MAC001

JOURNAL			
Date	Account	Debit	Credit
	<i>To apply overhead to Job MAC001</i>		

JOB COST SHEET
Dinosaur Vinyl, Inc.

Job No.: 5416
Customer: Macs & Cheese
Units Ordered: 1

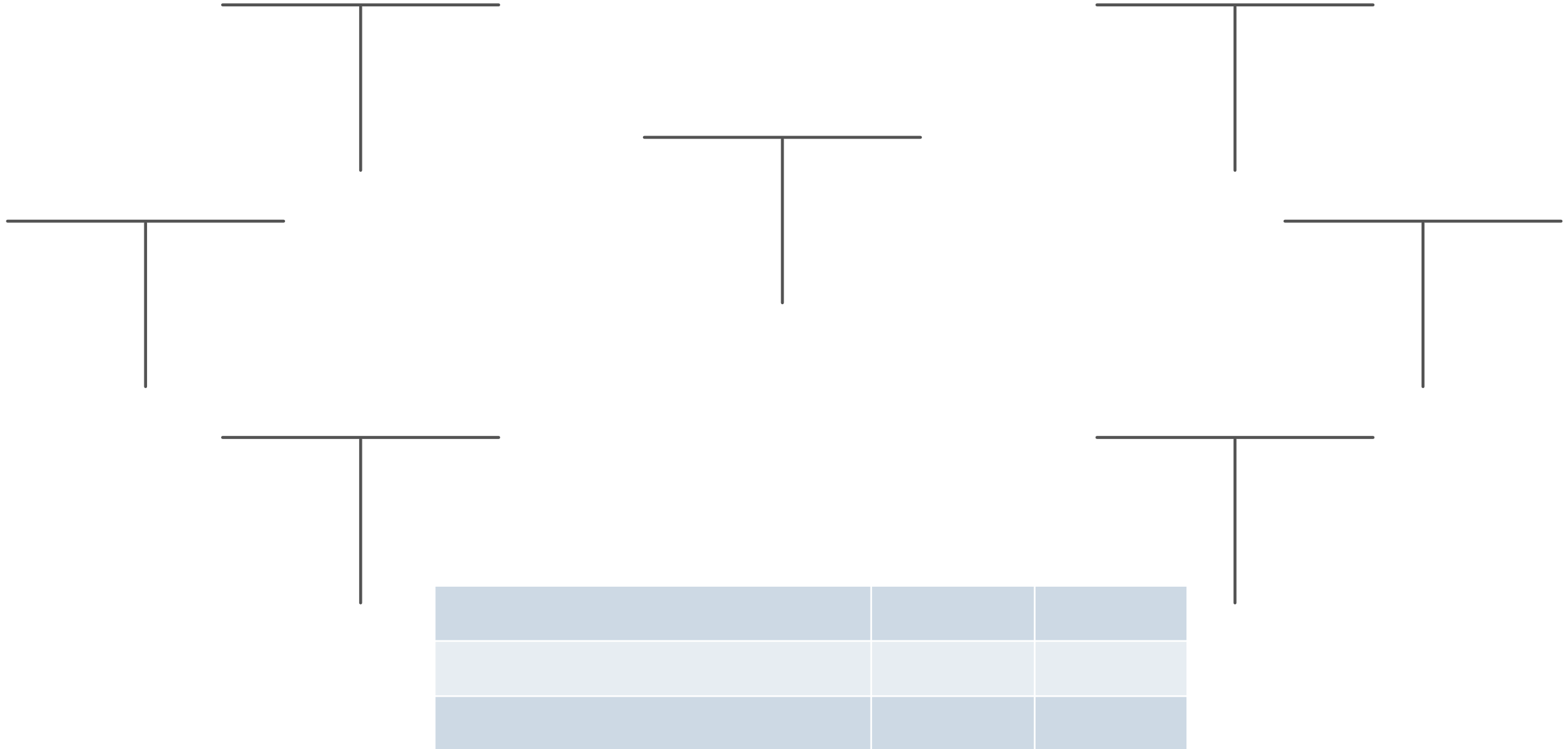
Customer No.: 2501723
Date Started: Dec. 22, 2018
Date Completed: Feb. 22, 2019

Direct Material	Units	Price	Amount
Vinyl	1	\$300	\$ 300
Black printing ink	2	50	100
Red printing ink	1	60	60
Gold printing ink	1	60	60
Grommets	12	10	120
Framing wood	40	1.5	60
Total Direct Materials			\$ 700
Direct Labor	Hours	Wage Rate	Amount
Material Handler	1	\$ 15	\$ 15
Print Technician	1	15	15
Production Assistants	2	18	36
Total Direct Labor			\$ 66
Manufacturing Overhead	Base Units	Rate	Amount
Direct Labor Cost			\$ 165
Total Manufacturing Overhead			\$ 165
Total Job Cost			\$ 931
Total Revenue			\$2,000

Example Overhead Item	Debit	Credit
Indirect Labor	Factory Overhead	Salaries Payable
Indirect Material	Factory Overhead	Inventory or Supplies
Factory Insurance	Factory Overhead	Prepaid Insurance
Factory Depreciation	Factory Overhead	Accumulated Depreciation
Taxes/Utilities	Factory Overhead	Taxes or Utilities Payable

06-30-X3	Factory Overhead		100,000	
	Salaries Payable			50,000
	Supplies			15,000
	Prepaid Insurance			5,000
	Accumulated Depreciation			11,000
	Taxes Payable			9,000
	Utilities Payable			10,000
	<i>To record various factory overhead costs</i>			

Manufacturing Overhead Allocation = Actual Overhead Incurred





**When MOH Applied (Allocated)
≠ Actual MOH**





Prepare Journal Entries for a Job Order Cost System

(Ending)

	Raw Materials Inventory: Vinyl	
Beginning inventory	1,000	
4/2/2017 (Purchases)	10,000	
Balance	11,000	

(Ending)

	Raw Materials Inventory: Black Ink	
Beginning inventory	300	
4/2/2017 (Purchases)	500	
Balance	800	

(Ending)

	Raw Materials Inventory: Red Ink	
Beginning inventory	300	
Balance	300	

(Ending)

	Raw Materials Inventory: Gold Ink	
Beginning inventory	300	
Balance	300	

	Raw Materials Inventory: Grommets	
Beginning inventory	120	
Balance	120	

	Raw Materials Inventory: Framing Wood	
Beginning inventory	60	
Balance	60	

JOURNAL			
Date	Account	Debit	Credit
	Raw Materials Inventory: Vinyl Raw Materials Inventory: Black ink Accounts Payable <i>To record purchase of vinyl and ink inventory</i>	10,000 500	10,500

JOURNAL			
Date	Account	Debit	Credit
	Work in Process Inventory Raw Materials Inventory: Vinyl Raw Materials Inventory: Black ink Raw Materials Inventory: Red ink Raw Materials Inventory: Gold ink Raw Materials Inventory: Grommets Raw Materials Inventory: Finishing wood <i>To record requisition of vinyl and ink inventory</i>	700	300 100 60 60 120 60

JOURNAL

Date	Account	Debit	Credit
	Work in Process Inventory (MAC001) Factory Wages Payable <i>To record labor for Job MAC001</i>	30	30

JOURNAL

Date	Account	Debit	Credit
	Work in Process Inventory Factory Wages Payable <i>To record labor for Job MAC001</i>	36	36

JOURNAL

Date	Account	Debit	Credit
	Work in Process Inventory Manufacturing Overhead <i>To apply overhead to Job MAC001</i>	165	165

JOURNAL			
Date	Account	Debit	Credit
	Finished Goods Inventory Work in Process Inventory <i>To recognize completion of Job MAC001</i>	931	931

JOURNAL			
Date	Account	Debit	Credit
	Cost of Goods Sold Finished Goods Inventory <i>To recognize sale of Job MAC001</i>	931	931
	Accounts Receivable Sales <i>To recognize sale of Job MAC001</i>	2,000	2,000

DINOSAUR VINYL, INC. Income Statement	
Sales	\$2,000
Cost of Goods Sold	<u>931</u>
Gross profit	\$1,069



In-class Exercises

In-class Exercise 1

THINK IT THROUGH

Allocating Costs

A manufacturing company has incurred these costs:

Purchase raw materials inventory	\$15,000
Issue raw materials inventory to Job A	3,000
Factory wage expense incurred	23,000
Factory wage allocated to Job A	2,000
Factory wage allocated to overhead	500
Manufacturing overhead incurred	7,500
Manufacturing overhead allocated to Job A	1,000

What is the cost allocated to Job A?

In-class Exercise 2: Tracking the Flow with Selected T-Accounts

A. Purchase raw materials inventory

B. Factory wage expense incurred

C. Issue raw materials inventory to Job P33

D. Factory wage allocated to Job P33

E. Factory wage allocated to overhead

F. Job P33 completed

G. Job P33 sold

Raw Materials Inventory: Vinyl

Work in Process Inventory

Factory Wage Expense

Finished Goods Inventory

Manufacturing Overhead

Cost of Goods Sold

Chapter 4, Problem Set A, Problem 13

The following events occurred during March for Ajax Company. Prepare a journal entry for each transaction.

Description	DR	CR
1. Received cash from customers on account.		
2. Paid cash for advertising.		
3. Received cash from customers on account.		
4. Paid cash for advertising.		
5. Received cash from customers on account.		
6. Paid cash for advertising.		
7. Received cash from customers on account.		
8. Paid cash for advertising.		
9. Received cash from customers on account.		
10. Paid cash for advertising.		
11. Received cash from customers on account.		
12. Paid cash for advertising.		

- A. Materials were purchased on account for \$35,429.
- B. Materials were requisitioned to begin work on Job C15 in the amount of \$25,259.
- C. Direct labor expense for Job C15 was \$24,129.
- D. Actual overhead was incurred on account of \$32,852.
- E. Factory overhead was charged to Job C15 at the rate of 200% of direct labor.
- F. Job C15 was transferred to finished goods at \$97,646.
- G. Job C15 was sold on account for \$401,000.

[illegible]

Chapter 4. PB2. Rulers Company is a neon sign company that estimated overhead will be \$60,000, consisting of 1,500 machine hours. The cost to make Job 416 is \$95 in neon, 15 hours of labor at \$13 per hour, and five machine hours. During the month, it incurs \$95 in indirect material cost, \$130 in other labor cost, \$320 in utilities, and \$350 in depreciation expense.

- A. What is the predetermined overhead rate if machine hours are considered the cost driver?
- B. What is the cost of Job 416?
- C. What is the factory overhead incurred during the month?

Chapter 4. PB3. Event Forms expects \$120,000 in overhead during the next year. It doesn't know whether it should apply overhead on the basis of its anticipated direct labor hours of 6,000 or its expected machine hours of 5,000. What would be the product cost under each predetermined allocation rate if the last job incurred \$3,500 in direct material cost, 55 direct labor hours, and 55 machine hours? Wages are paid at \$17 per hour.



Thank you!
Questions?